How Much Image Is Enough? Keeping Humans in the Loop of Image Download Infrastructure

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Abstract

Research infrastructures in the digital humanities increasingly hold documentation-grade images whose resolution far exceeds what most scholarly tasks require. Deep-zoom viewers make such images explorable in the browser, but researchers also need files they can keep, annotate, publish and teach with, and the original files are often too large to hand out safely or usefully. This paper describes how the Gothenburg Research Infrastructure in Digital Humanities (GRIDH) approached this problem by building a small on-demand download service that lets a person choose between three resolution levels of an image, generates the requested derivative only once, and stores it in project-scoped public storage. We present the design as a set of negotiations between what is safe and easy for the people who use the material and what is safe and efficient for the people who maintain the infrastructure. We argue that the question "how much image is enough?" is an interpretive, humanistic judgement that should stay with a human, and that infrastructures can be designed so that this judgement is visible, cheap to exercise and bounded in its consequences. We close with an account of the limitations of the current prototype.

Keywords

research infrastructure, image derivatives, IIIF, human in the loop, sustainability, cultural heritage data

1. Introduction

The phrase "human in the loop" usually refers to people who label, correct or approve the output of a machine learning system (Mosqueira-Rey et al. 2023). The theme of DHNB2027 invites a wider reading in which the humanities are part of the loops that make up technological systems in general. This paper takes that invitation to a place where loops are rarely discussed: the storage and delivery layer of a digital humanities research infrastructure. Nothing in the system we describe uses machine learning. It is nonetheless full of decisions about what a person may ask of a machine, what the machine may do on that person's behalf, and who carries the cost.

The Gothenburg Research Infrastructure in Digital Humanities (GRIDH) is a research node and infrastructure at the University of Gothenburg that creates and develops digital resources and publications for researchers and institutions (GRIDH 2026). Many of the projects on its portal revolve around images: paintings, inscriptions, rock art, archival documents. Depending on the project and on the purpose of the documentation, these images can be extremely large. A case we return to below is the documentation of Edvard Munch's monumental painting *Solen* (1911), which measures roughly 7.8 by 4.4 metres and is recorded in several image types, including a topographical capture of its surface.

For viewing, this is a solved problem. Each project exposes its images through the IIIF Image API (Appleby et al. 2020; Snydman, Sanderson, and Cramer 2015), so that a browser only ever fetches the tiles it needs. For *taking the image away*, it is not. Researchers, students and the interested public want a file: to put in a slide, to annotate in desktop software, to print, to cite, to compare side by side with something else. The original file is often the wrong answer to that request. It may be too large to download over an ordinary connection, too large to open on an ordinary laptop, and expensive for the infrastructure to serve repeatedly. Withholding it entirely is also the wrong answer, because openness is the point of the infrastructure.

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Our contribution is a description of the design reasoning behind a download service that sits between these two answers, and a reflection on what it means to keep humans, and humanistic judgement, in that loop. The service is a working prototype; we report on its design, not on a user evaluation.

2. Background

2.1. Infrastructure as a Relation

Star and Ruhleder argued that infrastructure is not a thing but a relation: something becomes infrastructure in relation to organised practices, and it becomes visible mostly when it breaks (Star and Ruhleder 1996; Star 1999). A download button is a good example. To a researcher it is a trivial affordance. To the unit that runs the server it is a commitment about bandwidth, disk, processing time and exposure to misuse. Decisions taken on one side of that relation are usually invisible on the other.

In digital humanities the two sides are often close. At GRIDH the people who build the infrastructure work alongside the researchers whose material it holds. That closeness is an asset: requirements can be negotiated between colleagues instead of being inferred from server logs.

2.2. Open Images and Their Costs

The case for releasing cultural heritage images openly and at high quality is well established (Terras 2015), and the FAIR principles ask that research data be accessible and reusable in practice, not only in principle (Wilkinson, Dumontier, Aalbersberg, et al. 2016). A file that exists but cannot reasonably be downloaded or opened is accessible only in a formal sense.

At the same time, openness has costs that fall on specific people and machines. Digital preservation has an environmental footprint, and Pendergrass and colleagues argue that practitioners should question the assumption that more copies, higher fidelity and permanent availability are always appropriate (Pendergrass et al. 2019). More recently, cultural heritage institutions have reported that automated collection of content for training generative AI models has put their online collections under heavy load (Weinberg 2025). Image servers that compute derivatives on request are particularly exposed, because each request can trigger real work. Padilla's call for responsible operations in libraries (Padilla 2019) applies here in a mundane sense: the operational layer is where responsibility is exercised or neglected.

Openness is also not the only value at stake. The CARE principles remind us that the people and communities connected to the material have interests that a purely technical notion of access does not capture (Carroll, Garba, Figueroa-Rodríguez, et al. 2020). Which projects offer downloads at all, and at what fidelity, is therefore a decision that belongs to each project.

3. Requirements as a Negotiation

We found it useful to state the requirements as two lists that pull against each other, and to treat the design as the place where they are reconciled.

Safe and easy for the people using the material. A person should be able to obtain an image file without knowing anything about tiles, pyramidal formats or API parameters. The choice they are offered should be expressed in terms they can reason about. The file they receive should open in ordinary software, in a common format. The same request should yield the same file at a stable address, so that it can be shared and cited. And the default should not leave them with a multi-gigabyte download they did not expect.

Safe and efficient for the people running the infrastructure. The service should never do the same expensive work twice. It should only fetch from sources the infrastructure itself controls. It should keep each project's files in that project's own space. It should survive two people asking for the same thing at the same moment. It should be small enough for a small team to understand, operate and repair.

Table 1
Resolution Levels Offered by the Download Service

Level	Label	Linear scale	Share of pixels	JPEG quality setting
1	high	100%	100%	95
2	medium	50%	25%	80
3	low	25%	6.25%	70

The single most consequential design decision follows from holding these lists together: *resolution is offered as a small, fixed set of levels rather than as a free parameter.* A free width-and-height parameter is more flexible, and it is what the IIIF Image API offers. But it gives an unbounded space of possible derivatives, which can neither be cached effectively nor explained simply. Three levels can be both.

4. The Download Service

4.1. Request and Resolution Levels

The service is a small HTTP application. A client sends a request naming the project's image API, the identifier of the image, a quality level and one or more output formats (JPEG or PNG). In a project's interface this corresponds to a person pressing a download control and picking a size.

The three levels are defined relative to the source image (Table 1). Level 1 keeps the full pixel dimensions. Level 2 halves each dimension and level 3 quarters it. Because area scales with the square of the linear dimension, level 2 contains a quarter of the pixels and level 3 about six per cent. The step from "full" to "half" is thus much larger in cost than its name suggests, and for many uses much smaller in perceptible loss.

For the topographical image of *Solen* mentioned above, the project API returns a record that points both to the published file and to the IIIF-ready source that backs the deep-zoom viewer. The service uses the IIIF source as its preferred starting point and falls back to the other files named in the record if that download fails. Reduced levels are produced with Lanczos resampling.

4.2. Workflow

Figure 1 shows the path of a request. The governing principle is that the cheapest check comes first. Before anything is fetched or computed, the service asks whether this exact derivative (project, image, level, format) already exists. If it does, the response is immediate and contains the address of the existing file. Only otherwise does the service retrieve the image record from the project API, download the source, convert it, write the result to storage and record it.

4.3. Project-Scoped Storage and Stable Addresses

GRIDH projects follow a naming convention in which each project lives at its own subdomain of the infrastructure's domain. The service derives the project name from the hostname of the API address and accepts it only if it matches that convention. The project name then determines where files are written: every project has its own folder in public storage, and within it a predictable path built from the image identifier, the level and the format. The file keeps the base name of the source file, so that what a person downloads is recognisably the same object they were looking at.

Two consequences matter for users. First, the address of a derivative is stable and can be shared. Second, each project's downloads form a browsable public directory that is independent of the service: if the service is switched off, the files it has produced remain available.

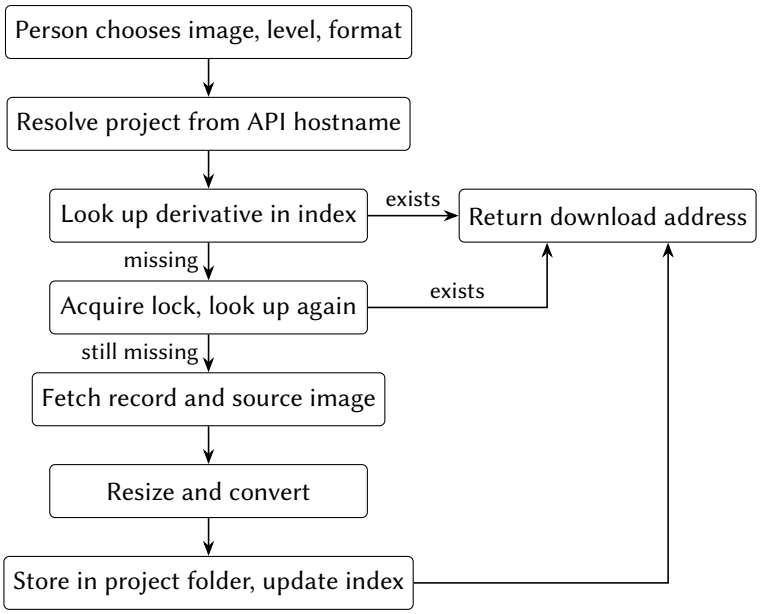


Figure 1: Path of a download request. A derivative that already exists is returned without contacting the project API or touching the source image.

4.4. Looking Up Without Searching

Checking whether a file exists sounds trivial, but repeatedly traversing directories becomes slow as the number of images grows. The service therefore keeps a lightweight index in an SQLite database, with one row per derivative keyed on project, image, level and format, together with a checksum of the file. The index is treated as a hint, not as the truth. When it reports a file, the service confirms that the file is really on disk. If it is not, the stale row is removed and the derivative is regenerated. An infrastructure maintained by a small team will sooner or later have someone tidy a directory by hand, and the system should recover from that without intervention.

4.5. Simultaneous Requests

When a project publishes a new image, several people may ask for the same derivative within seconds of each other. Without care, each request would see that the file is missing and start its own download and conversion. The service takes a lock specific to the image and level, and checks the index a second time once it holds the lock. The first request does the work; the others wait briefly and then receive the file the first one produced.

5. Discussion: Whose Loop?

5.1. The Researcher's Loop: Resolution as Interpretation

How much resolution is needed is not a technical question. It depends on what one wants to see. A lecture slide, a reading of brushwork, a comparison of compositions and a print for an exhibition call for different images of the same object. Deciding among them requires knowing the material and the argument one intends to make with it. This is exactly the kind of situated, interpretive judgement that the humanities are practised in, and it is the judgement that the service leaves with the person.

The alternative designs each remove that judgement. Always serving the original presumes that more data is better. Always serving a reduced image presumes that the infrastructure knows what the researcher needs. An adaptive system that guessed the appropriate size would hide the decision altogether. Three named levels make the decision explicit and make it cheap to revise: a person who

finds the medium image insufficient can ask for the full one, and the cost of having tried the smaller one first is negligible.

5.2. The Maintainer's Loop: Care and Limits

The second human in the loop is the research engineer. The design choices described above (compute once, verify the index against the disk, accept only known hosts, keep projects apart) are not optimisations for their own sake. They are what allows a small team to offer an open service without being afraid of it. Infrastructure that its maintainers do not trust tends to be closed down or locked away, and that is a worse outcome for users than a service with three sizes.

There is also an environmental argument. Each derivative that is reused instead of recomputed, and each download of a quarter of the pixels in place of all of them, is a small saving in energy and storage. Individually these are negligible. As a default embedded in every project's interface they add up, and they reflect the position that fidelity should be justified by use (Pendergrass et al. 2019).

5.3. The Project's Loop: Governance

The third loop is the project. The service is deliberately organised around projects, in storage layout, in naming and in the public directory each project receives. This mirrors how responsibility for the material is actually distributed. Whether a given collection should be downloadable at full resolution is a question for those who know its rights situation and the interests of the people connected to it (Carroll, Garba, Figueroa-Rodríguez, et al. 2020). A per-project structure does not answer that question, but it gives the answer somewhere to live.

5.4. Limitations

We want to be explicit about what the prototype does not yet do, because the tensions are instructive.

First, the service lets an operator state the project name explicitly when an API address does not follow the usual convention. This is convenient, and in the current implementation it also bypasses the hostname check. Convenience for the operator and safety for the infrastructure are in direct conflict here, and the check needs to apply regardless of how the project is named before the service is exposed more widely.

Second, the service currently places no upper bound on the pixel count of a source image, on the reasoning that sources come from the infrastructure's own storage and that documentation images are legitimately enormous. The source is also read into memory in full. Both are acceptable for trusted material at modest load and neither would be for arbitrary input.

Third, the lock that prevents duplicate work lives inside a single running process. It would not coordinate several server processes or machines.

Fourth, the service has no notion of who is asking. It does not authenticate, limit request rates, or distinguish a researcher from an automated harvester. The fixed set of levels bounds the work any requester can cause, to a small number of derivatives per image, but it does not address bulk collection of the results. Given current pressures on cultural heritage servers (Weinberg 2025), this is the question we consider most urgent.

Finally, we have not yet studied how researchers actually use the levels. Whether three is the right number, whether the labels are understood as intended, and how often the full-resolution level is really needed are empirical questions.

6. Conclusion

We have described a small piece of infrastructure and argued that its most important property is a division of labour between human and machine. The machine remembers what it has already made, keeps projects apart and refuses to do the same work twice. The human decides how much image is

enough. Designing for that division required stating openly that the needs of users and the needs of maintainers are different, and treating neither as a constraint to be minimised.

Read through the theme of this conference, the lesson is that keeping the humanities in the loop is not only a matter for systems that classify or generate. It is equally a matter for the unglamorous layers that decide what people are handed when they ask for a file. Future work will address the limitations listed above, connect the service to the download controls of GRIDH's image-based projects, and gather evidence on how the resolution levels are used in research and teaching.

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